Dr Moritz P. Schwarz

Curriculum Vitae

E moritz.schwarz@scmo.eu
W moritzschwarz.org

Education

DPhil (PhD), University of Oxford, UK. 2019–2023
Dissertation: *What Actually Happened? Novel Econometric Methods to Improve Estimates of Climate Impacts and Policies*.
Supervisors: Prof Sir David Hendry, Prof Cameron Hepburn. Clarendon Scholar.
Affiliated with Climate Econometrics at Nuffield College and INET Oxford.

MSc, University of Oxford, UK. 2016–2017
Environmental Change & Management. Graduated with Distinction.

BSc, London School of Economics, UK. 2013–2016
Environmental Policy with Economics. First Class Honours.

Academic Appointments

Einstein International Postdoctoral Fellow, 2023–
Faculty of Economics and Management, TU Berlin,
Potsdam Institute for Climate Impact Research (PIK), Germany.

Associate, University of Victoria, Canada. 2023–2025
SSHRC grant on extreme weather impacts in developing countries

Associate, INET Oxford, University of Oxford, UK. 2023–

Research Consultant, Columbia University, New York, USA. 2020–2021

Researcher / Associate, Climate Econometrics, Univ. of Oxford, UK. 2017–
Roles over time: Research Assistant, Doctoral Researcher, Associate Researcher (current).

Other Experience

Senior Climate Economist, Federal Ministry of Finance, Austria. 2021–
- Leading role in green budgeting framework and economic analysis; produced Austria's first empirical marginal abatement cost curve; macro-forecasting.
- Domestic climate and energy policy
- EU policy coordination on issues such as the EU ETS, CBAM, and the wider EU Climate Governance Architecture
- EU Issue Lead – Long-Term Finance & Financial Flows (UNFCCC); CBAM negotiation coordinator; negotiator on domestic climate and energy laws.

External Adviser, McKinsey & Co., UK/USA. Summer 2021

Austrian EU Presidency Coordinator, 2017–2019
Federal Ministry of Sustainability and Tourism, Austria.
- Coordinator for Austrian EU Presidency activities in international environmental governance
- EU negotiator in processes like the UNFCCC, CBD, UNEP/UNEA, IWC, etc

Research Interests

Environmental economics; climate & biodiversity policy; climate impacts; causal and reverse causal policy evaluation; time-series and panel econometrics; macro-forecasting.

References

Prof Cameron Hepburn — University of Oxford cameron.hepburn@smithschool.ox.ac.uk
Prof Robert Kaufmann — Boston University kaufmann@bu.edu
Prof Felix Pretis — University of Victoria fpretis@uvic.ca
Prof Linus Mattauch — TU Berlin linus.mattauch@tu-berlin.de
Prof Sir David Hendry — Nuffield College, Oxford david.hendry@nuffield.ox.ac.uk

Research and Publications

Job Market Paper

JMP

Pretis, F., and **Schwarz, M.**. Discovering What Mattered: Detecting Unknown Treatment as Breaks in Panel Models.

Abstract: Effective policy design requires knowing which interventions have made a measurable difference, yet outcomes such as emissions or growth often change without clear attribution. Rather than evaluating known policies, we start from the outcome itself and detect structural breaks in panel data, attributing them to potential causes. Formally, we develop an econometric framework that identifies unknown treatment timing and assignment as structural breaks in fixed-effects panel models. We show that a dummy-saturated Two-Way Fixed Effects model nests the standard staggered difference-in-differences design as a special case and that machine learning methods – such as indicator saturation and the adaptive LASSO – can recover treatment without prior knowledge. Established properties of indicator saturation allow control of the False Discovery Rate of discovered interventions. The approach provides a theoretical foundation for recent climate econometric applications and is demonstrated by identifying the Swedish carbon tax and detecting the economic impact of ETA terrorism. The methods are available in the open-source R package *getspanel*.

Peer-reviewed publications

2024

Mark, E., Rafaty, R., and **Schwarz, M.** (2024). Spatial-temporal dynamics of employment shocks in declining coal mining regions and potentialities of the 'just transition'. *Energy Policy* **195**: 114338.

2024

Stechemesser, A., Koch, N., Mark, E., Pretis, F., Ritter, N., **Schwarz, M.**, *et al.* (2024). Climate policies that achieved major emission reductions: Global evidence from two decades. *Science* **385**(6711): 884–892.
Selected media coverage: NYT, FT, CNN, WSJ, Economist, Le Monde, Spiegel, ORF.

Abstract: Meeting the Paris Agreement's climate targets necessitates better knowledge about which climate policies work in reducing emissions at the necessary scale. We provide a global, systematic ex post evaluation to identify policy combinations that have led to large emission reductions out of 1500 climate policies implemented between 1998 and 2022 across 41 countries from six continents. Our approach integrates a comprehensive climate policy database with a machine learning–based extension of the common difference-in-differences approach. We identified 63 successful policy interventions with total emission reductions between 0.6 billion and 1.8 billion metric tonnes CO_2. Our insights on effective but rarely studied policy combinations highlight the important role of price-based instruments in well-designed policy mixes and the policy efforts necessary for closing the emissions gap.

2024

Jiao, X., Pretis, F., and **Schwarz, M.** (2024). Testing for Coefficient Distortion due to Outliers with an Application to the Economic Impacts of Climate Change. *Journal of Econometrics* **239**(1).

2023

Thalheimer, L., **Schwarz, M.**, and Pretis, F. (2023). Large weather and conflict effects on internal displacement in Somalia with little evidence of feedback onto conflict. *Global Environmental Change*.

2022

Koch, N., Naumann, L., Pretis, F., Ritter, N., and **Schwarz, M.** (2022). Attributing agnostically detected large reductions in road CO_2 emissions to policy mixes. *Nature Energy* **7**: 844–853.
Selected media coverage: FT Sustainable Views, ORF, DER STANDARD, Salzburger Nachrichten.

2022

Scheer, A., Hopkins, D., **Schwarz, M.**, and Caldecott, B. (2022). Stranded labour in Alberta, Canada: A mixed methods study on the employment impacts of oil price in a resource-rich province. *Climate Policy* **22**(8): 1016–1032.

2022

Benoit, P., Clarke, A., **Schwarz, M.**, and Dibley, A. (2022). Decarbonization in state-owned power companies: Lessons from a comparative analysis. *Journal of Cleaner Production* **355**: 131796.

2020

Cohen, F., **Schwarz, M.**, Li, S., Lu, Y., and Jani, A. (2020). The challenge of using epidemiological case count data: The example of confirmed COVID-19 cases and the weather. *Environmental and Resource Economics: Perspectives*.

2018

Pretis, F., **Schwarz, M.**, Tang, K., Haustein, K., and Allen, M. R. (2018). Uncertain impacts on economic growth when stabilizing global temperatures at 1.5°C or 2°C warming. *Philosophical Transactions of the Royal Society A* **376**(2119): 20160460.
Selected media coverage: Carbon Brief, The Independent, Daily Mail, El Pais

R&R, Working Papers & Ongoing Research

Schwarz, M., Kurle, J., Pretis, F., Martinez, A., and Harper, G. (Cond. Accepted at *International Journal of Forecasting*). An Open-Source Empirical Macro (OSEM) Model for Economic and Emission Forecasting.

Schöngart, S., **Schwarz, M.**, Schwaab, J., Pretis, F., Pfleiderer, P., and Schleußner, C.-F. (R&R). Temporary warming, lasting losses: Economic damages from climate overshoot may be irreversible within this century.

Schwarz, M., and Pretis, F. (Working Paper). An empirical climate damage function accounting for climate extremes and adaptation. SSRN 4022690.

Schwarz, M. statIAM: Building a statistical emulator for Integrated Assessment Models.

Policy Papers and Reports

2025

Delgado, J., and **Schwarz, M.** (2025). Rückblick auf die 29. Weltklimakonferenz (COP29). *RWK – Reporting und Wirtschaft kompakt* 1/2025. Linde Verlag.

2025

Castle, J., Mark, E., Hendry, D. F., **Schwarz, M.**, and Pretis, F. (2025). Net zero: Strategy and tactics. *Newsletter of the Royal Economic Society*, Issue 208.

2024

Delgado, J., Haider, K., Lamprecht-Pühra, I., and **Schwarz, M.** (2024). Tracking von Klima- und Umweltschutzauszahlungen – der Bundesansatz. *SWK – Steuer und Wirtschaftskartei* **99**(6). Linde Verlag.

2022

Schwarz, M., Benoit, P., and Clark, A. (2022). Decarbonising state-owned power companies: A framework for applying policy actions. Smith School Working Paper No. 23-01.

2020

Cohen, F., Ives, M., Srivastav, S., **Schwarz, M.**, Lu, Y., Mealy, P., Bento Maffei De Souza, P., Jackson, P., and Hepburn, C. (2020). Emerging markets and self-interested climate action. A report prepared for Pictet AM by the Oxford Smith School of Enterprise and the Environment.

2020

Hepburn, C., and **Schwarz, M.** (2020). Climate change: Answers to common questions. Oxford Smith School of Enterprise and the Environment. A report prepared for Pictet AM.

2019

Mattauch, L., Creutzig, F., Franks, M., Funke, F., Jakob, M., Sager, L., **Schwarz, M.**, *et al.* (2019). Antworten auf zentrale Fragen zur Einführung von CO_2-Preisen. *Diskussionsbeiträge der Scientists for Future* **2**: 10.5281/zenodo.3371150.

2017

Mattauch, L., Roesti, M., **Schwarz, M.**, and Siegmeier, J. (2017). Wirtschaftswachstum und Klimawandel: Chancen und Herausforderungen auf dem Weg zur klimafreundlichen Gesellschaft. *Wirtschaftspolitische Blätter* **3**/2017.

Awards & Grants

2025: Oxford Policy Engagement Network Seed Fund Recipient.

2024: Best Paper Award for Junior Scholars, TU Berlin (Faculty VII).

2023: Einstein International Postdoctoral Fellowship, TU Berlin (~EUR 339k).

2023: British Academy Postdoctoral Fellowship offer (GBP 342k) – declined in favor of Einstein Fellowship.

2021: Insight Development Grant (Canada), co-recipient (CAD 60k).

2019–2023: Clarendon Scholarship, University of Oxford (~GBP 100k).

2017: ECI MSc Prize for Top Overall Performance & Best Dissertation, University of Oxford.

2016: George & Hilda Ormsby Prize (Best Undergraduate Dissertation), LSE.

Teaching

TU Berlin: Lecturer: Causal Inference (PhD level, design & instruction), Time Series (PhD level, design & instruction). Contributed to Environmental Economics teaching (MSc/BSc level).
Co-supervision of three BSc theses and two MSc thesis. Co-supervision of two PhD-students (leading supervision for empirical chapters).

University of Oxford: Lecturer: "New Environmental Economic Thinking" (MSc), TA: Environmental Economics (MSc), Quantitative Methods (MPhil).
Co-supervision of two MSc theses (both were subsequently published).

University of Nottingham: Designed & delivered R/tidyverse econometrics training (multi-day).

Peer Review

Peer review for Journals like: Econometrics Journal, International Journal of Forecasting, Economic Modelling, Nature Climate Change, Climate Policy, PNAS Nexus, Scientific Reports, Nature Communications, Energy Journal, Climatic Change, Economics of Disasters and Climate Change, PLOS One, Humanity and Social Sciences Communications, Frontiers in Climate - Mobility.

Policy & Public Outreach (selected)

Software Development: Major Contributor to the `gets` R-package, maintainer of the `getspanel` and the `osem` R-package. Over 159,000 total downloads.

Domestic Policy: Green budgeting, empirical MAC curve development, macro-forecasting. Research results featured in Austrian National Energy and Climate Plan. Policy exchanges with institutions including the German Chancellery, Ministry of Finance and Economics Ministry; Austrian Environment Agency; Climate Neutrality Forum (Brussels).

EU/UNFCCC: Issue Lead for Long-Term Finance; CBAM negotiation coordinator.

Coalition of Finance Ministers for Climate Action: Report on effectiveness of climate policies using results from the Science paper. Participation in the Green Macro Modelling Initiative with OSEM to improve policy modelling for policy-makers.

IPCC citations: Research prominently featured in SR1.5 and AR6.

Media: Coverage in major international outlets (CNN, WSJ, FT, NYT etc.).

Presentations (selected)

Conferences: EAERE (2018, 2024); Dynamic Econometrics (2024, 2026); VfS (2024); IIPF (2024); Int. Symposium on Forecasting (2024); EGU (2018, 2020, 2022, 2024); Econometric Models of Climate Change (2017, 2021, 2022, 2023, 2024, 2025, 2026); NOeG (2023, 2024).

Invited Seminars: Birkbeck London (2026); CEU Vienna (2025); IIEEE Lund (2025); University of Turin (2023).

Languages

German (native); English (C2); French (conversational).